# DynSGS setups of the paper's numerical tests

decks of branch `sm/MHD` at commit `2bb4c2a1`; notation of the paper, (11)–(16) and Table *split*

| | **Smooth vortex, MHD** | **Smooth vortex, Euler**[b] | **Brio–Wu**[a] | **Orszag–Tang** | **Rotor** | **Blast** | **Jet, Mach 800** | **Flux emergence**[c] |
|---|---|---|---|---|---|---|---|---|
| *deck in* `problems/MHD/` *unless marked* | `smoothVortex` | `CompEuler/ smoothVortex` | `brioWu1d` | `orszagTang-Bormanis2024` | `rotorDaoNazarov2022` | `blastBalsara-Spicer1999` | `astroJetWuShu2018` | `fluxEmergence-Son2025DSGS` |
| **Discretization (deck; paper, where its text or captions say otherwise)** | | | | | | | | |
| Dimension, equations | 2D, 9 | 2D, 4 | 1D, 8 (no $\psi$) | 2D, 9 | 2D, 9 | 2D, 9 | 2D, 9 | 2D, 9 |
| $\gamma$ (`:dsgs_gamma`) | 5/3 | 1.4 | 2 | 5/3 | 1.4 | 1.4 | 1.4 | 1.05 |
| $N$ | 4 (figure: a sweep)[d] | 4 (figure: a sweep)[d] | 4; paper 4, 5, 6, 7 | 4 | 3; paper 4 | 3; paper 4 | 4 | 4 |
| Elements | $16^2$ on $[-10,10]^2$ | $16^2$ on $[-10,10]^2$ | $600/N$: 150, 120, 100, 86 | $64^2$ ($32^2$, refined once); paper $128^2$ to $512^2$ | $128^2$ ($32^2$, refined twice); paper $256^2$ | $64^2$ ($32^2$, refined once); paper $128^2$, $256^2$ | $80 \times 240$ ($20 \times 60$, refined twice), half domain | $160 \times 70$ ($80 \times 35$, refined once) |
| $\Delta t$ | $2 \cdot 10^{-3} \cdot 64/(n_{el}N)$ | $2 \cdot 10^{-3} \cdot 64/(n_{el}N)$ | $5 \cdot 10^{-5} \cdot 600/\mathrm{DOFs}$ | $2 \cdot 10^{-4}$; paper $10^{-4}$ to $2.5 \cdot 10^{-5}$ | $2 \cdot 10^{-4}$ | $10^{-5}$ | $1.25 \cdot 10^{-7}$ | $5 \cdot 10^{-3}$ |
| $t_{\mathrm{end}}$ | 1 | 1 | 0.1 | 1 | 0.15 | 0.01 | $2 \cdot 10^{-3}$ | 50; paper figure 51 $\tau_0$ |
| Time integrator | `CarpenterKennedy2N54` at fixed $\Delta t$ in every deck, as in the paper | | | | | | | |
| **The coefficient: (11)–(16)** | | | | | | | | |
| Sensor $R_i$ (`:dsgs_sensor`) | (R) residual | (R) residual | (R) residual | (L) legacy | (L) legacy | (L) legacy | (R) residual | (L) legacy |
| $\mu\|_e$ computed (`:dsgs_freeze_stage`) | once per step | once per step | once per step | every RK stage | every RK stage | every RK stage | once per step | every RK stage |
| $C_1$, $C_2$ (`:dsgs_CR`, `:dsgs_Cmax`) | 1, 0.5 | 1, 0.5 | 1, 0.5 | 1, 0.5 | 1, 0.5 | 1, 0.5 | 1, 0.5 | 1, 0.5 |
| Floor $C_{\min}\Delta_e(\|\boldsymbol{u}\| + c_f)$ (`:dsgs_Cmin`) | 0 | 0 | 0.06 | 0 | 0 | 0 | 0 | 0.03 |
| Length $\Delta_e$, (15) | $h_e/(N+1)$ | $h_e/(N+1)$ | $h_e/(N+1)$[a] | $h_e/(N+1)$ | $h_e/(N+1)$ | $h_e/(N+1)$ | $h_e/(N+1)$ | $h_e/(N+1)$ |
| Cutoff $c$ (`:dsgs_cutoff`) | 0 in every deck (no cutoff; the Alfvén wave, not in the paper, runs $10^{-3}$) | | | | | | | |
| $\mu\|_e = 0$ for the first (steps) (`:dsgs_hold_steps`) | 2 | 2 | 2 | 2 | 2 | 0 | 0 | 2 |
| Residual set $\mathcal{I}$ | 8, $\psi$ excluded | $\rho$, $\|\boldsymbol{m}\|$, $E$ | all 8 | 8, $\psi$ excluded | 8, $\psi$ excluded | 8, $\psi$ excluded, $+\rho e$ (`:dsgs_thermal_residual`) | 8, $\psi$ excluded | 8, $\psi$ excluded |
| Denominator of (12) (`:dsgs_norms`) | domain, $\Omega$ | domain, $\Omega$ | domain, $\Omega$ | domain, $\Omega$ | element, $\Omega_e$ | element, $\Omega_e$ | element, $\Omega_e$ | element, $\Omega_e$ |
| Floors $\sigma_i$, (16) (`:dsgs_rel`, `:dsgs_local_rel`, `:dsgs_fast_floors`) | $\epsilon_{\mathrm{rel}} = 1$: $\bar{\rho}$, $\bar{\rho}\bar{c}$, $\bar{\rho}\bar{c}^2$, $\sqrt{\bar{\rho}}\,\bar{c}$ | $\epsilon_{\mathrm{rel}} = 1$: $\bar{\rho}$, $\bar{\rho}\bar{c}$, $\bar{\rho}\bar{c}^2$ | $\epsilon_{\mathrm{rel}} = 1$, as SV | $\epsilon_{\mathrm{rel}} = 1$, as SV | element scales $\rho_e$, $\rho_e c_e$, $\rho_e c_e^2$, $\sqrt{\rho_e}c_e$, $\times 1$ | element scales, $c_e \to c_{f,e}$ | element scales, $c_e \to c_{f,e}$ | element scales, $\times 1$ |
| Speed in the cap (13) | $\|\boldsymbol{u}\| + c_f^{\max}$ | $\rho\,(\|\boldsymbol{u}\| + c)$, dynamic[b] | $\|\boldsymbol{u}\| + c_f$, exact $x$-fast speed | $\|\boldsymbol{u}\| + c_f^{\max}$ | $\|\boldsymbol{u}\| + c_f^{\max}$ | $\|\boldsymbol{u}\| + c_f^{\max}$ | $\|\boldsymbol{u}\| + c_f^{\max}$ | $\|\boldsymbol{u}\| + c_f^{\max}$ |
| **Distribution over the equations: the paper's Table *split* (coefficients of $\nabla$ of the diffused variable; $\nu = \mu\|_e$)** | | | | | | | | |
| Diffused variables (`:dsgs_conserved`, `:dsgs_ref_weight`) | physical: $\rho, \boldsymbol{u}, T, \boldsymbol{B}, \psi$ | physical: $\rho, \boldsymbol{u}, T$ | conserved $\boldsymbol{q}$ | physical: $\rho, \boldsymbol{u}, T, \boldsymbol{B}, \psi$ | conserved $\boldsymbol{q}$ | conserved $\boldsymbol{q}$ | conserved $\boldsymbol{q}$ | $(\boldsymbol{q} - \boldsymbol{q}_e)/\rho_e$; $\boldsymbol{B}, \psi$: $\boldsymbol{q} - \boldsymbol{q}_e$ |
| $\lambda_i$ (`:μ`) | 1 for every equation in every deck, so $\lambda_1 = 1$: mass diffusion is on (the paper: $\lambda_1 = 0$) | | | | | | | |
| $\rho$ | $\nu$ | $\nu$ ($\mu/\|\rho\|_{\infty,e}$) | $\nu$ | $\nu$ | $\nu$ | $\nu$ | $\nu$ | $\rho_e\nu$ |
| $\rho u, \rho v, \rho w$ | $\bar{\rho}_e\nu$, stress $\boldsymbol{\tau}$ | $\mu$, stress $\boldsymbol{\tau}$ | $\nu$ | $\bar{\rho}_e\nu$, stress $\boldsymbol{\tau}$ | $\nu$ | $\nu$ | $\nu$ | $\rho_e\nu$ |
| $E$ (`:dsgs_nazarov_energy`) | $\bar{\rho}_e\nu/\mathrm{Pr}_t$ on $\nabla T$; $+\boldsymbol{\tau}\cdot\boldsymbol{u}$, $+\eta\boldsymbol{B}\cdot\nabla\boldsymbol{B}$ | $\mathrm{P}\,\mu/(\gamma-1)$ on $\nabla T$; $+\boldsymbol{\tau}\cdot\boldsymbol{u}$ | $\nu$ on $E$ | $\bar{\rho}_e\nu/\mathrm{Pr}_t$ on $\nabla T$; $+\boldsymbol{\tau}\cdot\boldsymbol{u}$, $+\eta\boldsymbol{B}\cdot\nabla\boldsymbol{B}$ | $\nu$ on $E$ | $\nu$ on $E$ | $\nu$ on $E$ | thermal part: $\rho_e \max(c_E\nu_c, \nu_{\mathrm{fl}})$, $c_E = \gamma(\gamma-1)/\mathrm{Pr}_t$; rest: $\rho_e\nu$ |
| $\mathrm{Pr}_t$ (`:dsgs_Prt`) | 1 | P = 0.1 (`:Pr`)[b] | 1 (not used) | 1 | 1 (not used) | 1 (not used) | 1 (not used) | 1 |
| $B_x, B_y, B_z$; $\psi$ | $\eta = \nu$; $\nu$ | — | $\nu$; — | $\eta = \nu$; $\nu$ | $\nu$; $\nu$ | $\nu$; $\nu$ | $\nu$; $\nu$ | $\nu$; $\nu$ |
| **Positivity (not part of DynSGS)** | | | | | | | | |
| State limiter (`:positivity_method`) | none | node repair; $10^{-6}$ of the core $\rho$, $p$ | none | none | conservative; $\rho, p \geq 10^{-6}$ | conservative; $10^{-6}$, $10^{-7}$ | conservative; $1.4 \cdot 10^{-7}$, $10^{-6}$ | none |

Red: differs from the paper's text. (R), (L): the residual and legacy sensors of `dsgsCutoff.pdf` — (L) is (11) at every RK stage against the assembled $M^{-1}\mathcal{R}$, (R) is (11) once per step against the element's own $M_e^{-1}\mathcal{R}_e$. $c_f^{\max} = \sqrt{(\gamma p + |\boldsymbol{B}|^2)/\rho}$, the fast speed bounded over directions. $\nu_c$: the residual coefficient before the $C_{\min}$ floor $\nu_{\mathrm{fl}}$. $T = p/\rho$. Unset keys take `src/io/mod_inputs.jl`'s defaults; environment overrides are taken unset. Notes *a*–*d*: next page.

## Notes to the table

*a* Since `2bb4c2a1` (2026-10-19) the Brio–Wu deck uses the paper's $\Delta_e = h_e/(N+1)$ with $(C_1, C_2, C_{\min}) = (1, 0.5, 0.06)$, as every other deck. Before, it scaled them by $((N+1)/N)^2$, $(N+1)/N$, $(N+1)/N$, i.e. $(1, 0.5, 0.06)$ on Dao & Nazarov's $\Delta_e = h_e/N$, the same at every $N$ for a fixed number of DOFs; `JEXPRESSO_BW_HSCALE=nop` still gives that. The paper varies $C_{\min}$ (`JEXPRESSO_BW_CMIN`).

*b* The Euler vortex uses the CompEuler kernel: $\mu_{\text{res}}|_e = C_1 \Delta_e^2\, n_\rho\, r_e$ and $\mu_{\max}|_e = C_2 \Delta_e \|\rho\|_{\infty,\Omega_e} \|\|\boldsymbol{u}\| + c\|_{\infty,\Omega_e}$ are dynamic ($n_\rho$: the density denominator); the momentum residual enters as one vector norm; heat conduction $\mathrm{P}\mu/(\gamma-1)$ with the artificial Prandtl number $\mathrm{P} =$ `:Pr` $= 0.1$ (default). The deck's `:dsgs_Prt => 0.7` is not read by this kernel.

*c* The paper's $160 \times 70$ run is `fluxEmergenceSon2025DSGS`. Its sibling `fluxEmergenceSon2025` ($80 \times 35$, $\Delta t = 7.5 \cdot 10^{-3}$, $t_{\text{end}} = 54$, a case positivity floor as stage limiter) diffuses $\boldsymbol{q} - \boldsymbol{q}_e$ with $\nu$ (no $\rho_e$ weight) and is otherwise the same.

*d* The deck's default is one point of the convergence figure; `JEXPRESSO_SV_NOP`, `_NELX` (`JEXPRESSO_EV_*` for the Euler vortex) set the others, with $\Delta t$ from the rule.

## Where the paper's text and the decks differ

These are the red entries of the table, with what each one changes.

1. **Sensor, (11).** The paper's (11) reads the assembled rate $M^{-1}\mathcal{R}$, which is the legacy sensor (L) when it is evaluated at every stage. The smooth vortices, Brio–Wu and the jet run (R): the element's own rate $M_e^{-1}\mathcal{R}_e$, once per step. Orszag–Tang, the rotor, the blast and the flux emergence run (L).
2. **Mass diffusion.** The paper sets $\lambda_1 = 0$ (mass untouched); every deck has `:μ[1] = 1`, so $\nu\nabla\rho$ is on, in all eight setups.
3. **Energy.** The paper's Table *split* has $\lambda_4 \bar{\rho}_e \mu|_e\, \gamma/((\gamma-1)\mathrm{Pr}_t)$ on $\nabla T$ with $\mathrm{Pr}_t = 0.7$. No deck uses that combination: the physical-form decks (the MHD vortex, Orszag–Tang) take Dao & Nazarov's $\bar{\rho}_e \nu/\mathrm{Pr}_t$ (`:dsgs_nazarov_energy`) with $\mathrm{Pr}_t = 1$, and the conserved-form decks diffuse $E$ itself (the flux emergence, its thermal part only).
4. **Ohmic heating.** The paper's remark says the resistive term is not returned to the energy. The physical form (the MHD vortex, Orszag–Tang) adds the resistive work $\nabla \cdot (\eta \boldsymbol{B} \cdot \nabla \boldsymbol{B})$ to the energy, so the magnetic energy removed reappears as heat and the total energy is conserved.
5. **Conserved form.** The paper's stabilized system (stress, $\kappa_e \nabla T$, $\boldsymbol{\tau} \cdot \boldsymbol{u}$) is the physical form. Brio–Wu, the rotor, the blast, the jet and the flux emergence diffuse the conserved variables with one kinematic $\nu$ instead: no stress, no work terms.
6. **Normalization, (12) and (16).** (12) normalizes by the domain spread; the rotor, the blast, the jet and the flux emergence normalize by the element's own spread, floored at the element's scales. (16) has $\epsilon_{\text{rel}} = 10^{-3}$; every deck runs $\epsilon_{\text{rel}} = 1$, the floor being the full physical scale.
7. **Constants.** $C_{\min}$ is not in (12)–(14); Brio–Wu runs 0.06 and the flux emergence 0.03.
8. **Per stage or per step.** The paper's algorithm evaluates DynSGS once per right-hand-side call (every stage); the (R) decks compute $\mu|_e$ once per step.
9. **Not in the paper.** The start-up hold (2 steps; none in the blast and the jet), the fast-speed floors (blast, jet), the residual of $\rho e$ (blast) and the reference weight $\rho_e$ (flux emergence).
10. **Discretization.** The rotor (paper $256^2$ at $N = 4$; deck $128^2$ at $N = 3$), the blast (paper $128^2$, $256^2$ at $N = 4$; deck $64^2$ at $N = 3$) and Orszag–Tang (paper $128^2$ to $512^2$ in the figures and $32^2$ with $\Delta t = 5 \cdot 10^{-4}$ in the text; deck $64^2$, $\Delta t = 2 \cdot 10^{-4}$) were run on other meshes than the decks'. The flux-emergence figure is at $t = 51\,\tau_0$; the deck stops at 50.

## Defaults that changed while the paper's runs were made

A run made before these dates used the old behaviour, whatever its deck says now.

- **2026-09-10** (`39362212`): the residual sensor (R) becomes the default and `:dsgs_sensor => "legacy"` keeps the old one. Before, every DynSGS run used (L).
- **2026-09-23** (`8a81279c`): the floors (16) go from $\epsilon_{\text{rel}} = 10^{-3}$ to $\epsilon_{\text{rel}} = 1$ (`:dsgs_rel`). Before, a nearly uniform field (the density of a vortex, $B_x$ in a shock tube) had its residual divided by $10^{-3}$ of its scale.
- **2026-10-14** (`b956ccd` for `DSGS_MHD`, `e12fca8` for the CompEuler `DSGS`): with the residual sensor, $\mu|_e$ is computed once per step. Before, at every stage, where the residual stalls at $O(\Delta t)$ on smooth flow (`dsgsCutoff.pdf`).
- **2026-10-19** (`2bb4c2a1`): Brio–Wu moves from $\Delta_e = h_e/N$ to the paper's $h_e/(N+1)$ (note *a*).
- **Last change of each deck:** Brio–Wu `2bb4c2a1` (2026-10-19), MHD vortex `b956ccd` (2026-10-14), Euler vortex `e12fca8` (2026-10-14), Orszag–Tang `6829a2c` (10-08), rotor `7bd7bc2` (10-09), blast `e4fcf14` (10-12), jet `2590be6` (10-12), flux emergence `c1d5b46` (10-13).